

IMMIDI SHREYAS

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PROFILE

Full-stack Software Engineer building **AI/LLM products** deployed in live **semiconductor manufacturing** environments, hosted on **AWS** end-to-end. Delivered a shop-floor diagnostics platform that cut client troubleshooting costs by up to **50%** and let junior operators match senior-expert resolution speed in **8/10** blind test cases. **AWS Certified Solutions Architect – Associate**. Skilled across **Python, React.js, Node.js, Java, and AWS**.

KEY SKILLS

Cloud (AWS): EC2, S3, ECS, Lambda, CloudFormation, IAM — AWS Certified Solutions Architect – Associate

Languages & Frameworks: Python, Java (Spring Boot), JavaScript/TypeScript, React.js, Node.js

AI / ML: RAG pipelines, agentic AI, multimodal LLMs, vision-based document understanding, ChromaDB (vector search), time-series forecasting

DevOps & Data: Docker, Jenkins CI/CD, Terraform, PostgreSQL, QuestDB, MongoDB

EXPERIENCE

Software Engineer — ASM Technologies

Aug 2025 – Present

AI diagnostics platform for field engineers & operators in semiconductor manufacturing

Stack: Python (backend), React.js (frontend), AWS, QuestDB, PostgreSQL, ChromaDB (vector store), Endee

- Built a **log analyzer** and **Live Recipe** engine that correlate equipment logs, live machine output, and technical documentation to give operators specific root-cause fixes for machine bring-up, downtime, and process-drift issues — driving up to **50% reduction** in client troubleshooting cost.
- Ran blind test cases pairing junior (L1) operators using the tool against unaided senior (L3) experts: **L1s matched or beat L3 resolution time in 8 of 10 cases**, cutting dependence on senior experts who are scarce and frequently off-site when issues occur.
- Built **DiagramGPT**, a document-intelligence feature that reads image-heavy manuals and CAD files (.dwg, .sld) alongside text, so operators can query diagrams and schematics directly instead of text-only instructions.
- Created an **agentic AI system for reading electrical schematics**, achieving **91% accuracy** in identifying components and interpreting circuit logic to support troubleshooting.
- AWS infrastructure & architecture:** designed and deployed the platform on **AWS (EC2, S3, ECS, CloudFormation)**, applying AWS Certified Solutions Architect – Associate principles for secure, cost-optimized, highly available production deployments across client environments.
- Delivered as a configurable B2B product, adapting knowledge base and workflows to each client's equipment fleet and documentation.

Web Development Intern — Tech Mahindra

Feb 2025 – May 2025

- Built microservices with Spring Boot, PostgreSQL, and the Redfish API for an internal firmware-tracking platform.

Intern — L&T Finance

Aug 2023 – Sep 2023

- Supported backend services and OCR-based PAN card verification in Flutter apps during Agile sprints.

PROJECTS

Day-Ahead Electricity Price Spike Forecasting & Hedging Simulation

July 2026 - Present

Python, LightGBM, SARIMAX, Time-Series Forecasting, pandas, scikit-learn

- Built an end-to-end forecasting pipeline on day-ahead electricity price and grid load data, engineering **13 features** (lagged prices, rolling weather aggregates, renewable supply proxies) across **3 grid locations**.
- Trained a gradient-boosted model (LightGBM) achieving **3.9% lower MAE** (4.93 vs. 5.13 EUR/MWh) than a SARIMAX baseline, validated via walk-forward backtesting across multiple independent time windows.
- Designed a spike-detection classifier using an adaptive rolling-90th-percentile threshold, achieving **94.2% recall** on abnormal price events occurring at a ~10.6% base rate.
- Simulated a model-guided hedging strategy against a no-hedge baseline, demonstrating **€2,875 in avoided cost** across a ~17,500-step backtest under stated cost assumptions.

Music Streaming Platform

Jun 2024 – Aug 2024

MERN stack: MongoDB, Express.js, React.js, Node.js, Docker, AWS ECS

- Designed RESTful APIs with **JWT-based authentication**; architected and deployed microservices in **Docker containers on AWS ECS**.
- Optimized MongoDB schema design, improving music-library query performance by **15%**.
- Built a responsive React.js frontend consuming the Node/Express APIs, with client-side caching to cut redundant network calls, and set up a Jenkins CI/CD pipeline for automated build and deployment.

GitHub Profile Summarizer

May 2024 – Jul 2024

Python, Machine Learning, NLP

- Built a Python web scraper (BeautifulSoup + GitHub REST API) that aggregates repository and contribution history into structured JSON.
- Used **Meta Llama** to auto-generate candidate summaries from repo activity, normalizing signals (languages used, commit patterns, README quality) into consistent, comparable profiles.

EDUCATION & CERTIFICATIONS

B.Tech, Computer Science — Vellore Institute of Technology, Chennai (CGPA 8.4) | May 2021 – May 2025

AWS Certified Solutions Architect – Associate

MERN Full Stack (ETHNUS) • AI/ML (Google Developers & Smart Internz)

LANGUAGES

English, Hindi, Telugu (Fluent), Kannada (Native), German (Basic)